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[5152]-168

S.E. (Comp. Engg.) (II Sem.) EXAMINATION, 2017
MICROPROCESSOR AND INTERFACING TECHNIQUES
(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Answer total *four* questions Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4, Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

1. (a) Draw architecture of 80386DX processor with neat sketch. List and explain different types of 80386 processors [6]
(b) Write a short note on TSR. [3]
(c) Explain different directives used within 80386 programming. List difference between Near and Far procedure. [3]

Or

2. (a) Draw the skeleton for .COM and .EXE file. Explain various terms related to .COM and .EXE file. [4]
(b) Write initialization instruction for 8259A to meet the following specifications : [4]
(i) Single level triggered mode
(ii) Call address interval of 4
(iii) Non-buffered
(iv) No special fully nested mode.

P.T.O.

- (c) Explain the following 80386 addressing modes with appropriate examples : [4]
- (i) Scaled Indexed Mode
 - (ii) Based Scaled Indexed Mode
 - (iii) Based Scaled Indexed with Displacement Mode
 - (iv) Direct Addressing Mode.
3. (a) Explain in detail different operating modes of 8251A. [6]
- (b) Explain R to 2R ladder network with respect to DAC. [3]
- (c) Explain the following control signals with respect to 8255 Mode1 (Strobed I/O Mode) [3]
- (i) $\overline{STB}$
 - (ii) IBF
 - (iii) $\overline{OBF}$.

Or

4. (a) Draw and explain architecture diagram for 8279. [6]
- (b) What are the different types of Data Communication ? List difference between synchronous and asynchronous communication. [3]
- (c) Why DMA controlled data transfers are preferred over interrupt driven or program controlled ? Explain the use of HRQ and EOP signals. [3]

5. (a) Draw the schematic of 8086 microprocessor operating maximum mode, showing supporting chips required like clock generator, latches, buffers etc. Explain working of 8086 in Maximum mode. [7]
- (b) Explain the following instruction of 8087 with example : [3]
- (i) FSQRT
- (ii) FPTAN
- (iii) FLDPI
- (c) Define the following terms : [3]
- (i) Accuracy
- (ii) Monotonicity
- (iii) Offset error.
- Or
6. (a) Draw and explain the read cycle timing diagram of 8086 in minimum mode. [6]
- (b) Draw and discuss the interface and interaction between 8086 and 8087. [7]
7. (a) List and explain the features of Intel X58 Chipset. [7]
- (b) Explain the features of 82801 I/O controller hub. [4]
- (c) Draw basic blocks of Intel X58 chipset. [2]

Or

8. (a) Draw and explain block diagram of i5 motherboard. [7]
- (b) Discuss the following signals of Power Management interface of 82801IJR I/O controller hub : [6]
- (i) PLTRST#
 - (ii) THRM#
 - (iii) THRMTRIP#
 - (iv) PWROK.